Lesson plan · for the teacher

One neuron

AI · Level 3 · Lesson 1

Time
40 min

Materials
Board and chalk; Three input cards per group, marked 1 on one side and 0 on the other; Weight cards (3, 2, 1) and a bias card (-1) for each group

Learning goals

- Work out a neuron's output from its inputs, weights and bias.
- Explain weights as how much each input matters, and learning as changing them.
- Explain why one neuron can only draw one straight line between two answers.

Activity: Be a human neuron

1. Warm-up (5 min): Ask: how do you decide whether to take an umbrella in the morning? List the clues pupils use (clouds, the forecast, the season) and ask which one they trust most.
2. Teach (8 min): Draw three input circles joined to one neuron. Write 1 or 0 in each input and a weight on each line (forecast 3, season 2, clouds 1). Work out one day aloud: multiply, add, add the bias (-1), and say yes if the total is above zero.
3. Human neuron (14 min): In groups of four, three pupils hold the input cards and the fourth is the neuron. Call out a day ("cloudy, no rain forecast, rainy season"); input pupils show 1 or 0, and the neuron works out the total and decides. Play five days and record the answers.
4. Change a weight (7 min): Set the clouds' weight to 0 and replay one day: what changed? Then make the forecast's weight -3: what does a negative weight do? Ask which setting gives the most sensible answers.
5. Wrap-up (6 min): Ask: can our neuron decide "the classroom is comfortable: not too cold, not too hot"? Draw a temperature line and show that one cut can't isolate the middle. Then give the exit ticket.

Make input cards (1 on one side, 0 on the other) and weight cards (3, 2, 1) and a bias card (-1) for each group. Plan five days to call out, including one where the neuron gets it wrong.

Discussion

- Which input in our umbrella neuron should matter most, and why?
- What would a weight of zero mean in real life?
- Which decisions in school are "just right in the middle"?

Exit ticket

- Inputs 1, 0 and 1, weights 1, 3 and 2, bias -4: what is the total, and what does the neuron say?

Try it at school tomorrow
Play "Human neuron": three pupils hold input cards for the umbrella question (cloudy? rain forecast? rainy season?), each showing 1 or 0, and each card has a weight written on it (3 for the forecast, 2 for the season, 1 for the clouds). A fourth pupil multiplies each card by its weight, adds the bias (-1) and decides: umbrella or not? Then make one weight zero, then negative, and decide again.

Practice questions · for you

One neuron

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1 **Which of these can one neuron decide on its own?**

- ◯ The bus is on time: neither early nor late
- ◯ Pass: the mark is above 50
- ◯ Soil is just right: neither too dry nor too wet

2 **Work out the neuron's answer.**

Inputs 1, 0 and 1 have weights 2, 3 and 1, and the bias is -2. The total is _____, so the neuron says _____.

Word bank: yes · no · 1 · 3

3 **If the umbrella neuron's bias changes from –1 to –5, it will say "take the umbrella" more often.**

◯ True ◯ False

4 **Match each part of a neuron to its place in a school marking scheme.**

Inputs	• • • •	"The exam counts three times as much as homework"
Weights	• • • •	A pupil's marks for the exam and the homework
Bias	• • • •	Pass or fail
Output	• • • •	The pass mark the total must beat

✎ **Think of a yes-or-no decision in your school. Which three inputs would a neuron need, and which would get the biggest weight?**

Check your answers

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1 **Pass: the mark is above 50**

"Above 50" has one cut: yes on one side, no on the other, so one straight line does it.

2 **Inputs 1, 0 and 1 have weights 2, 3 and 1, and the bias is −2. The total is 1, so the neuron says yes.**

1 × 2 + 0 × 3 + 1 × 1 = 3, and the bias of −2 brings it to 1.

3 **False**

The bias is added to every total. A more negative bias lowers every total, so fewer days get above zero: the neuron becomes more reluctant to say yes.

4 **Inputs — A pupil's marks for the exam and the homework**
Weights — "The exam counts three times as much as homework"
Bias — The pass mark the total must beat
Output — Pass or fail

A marking scheme works like a neuron: marks (inputs) are weighted and added, the total is compared with a threshold that the bias sets (a pass mark of 50 is a bias of −50), and the answer is yes or no.